

Nishant Suthar — Full Portfolio & Project Reference

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This document is a comprehensive reference covering all projects, education, certifications, and experience. It is intended for detailed reference use (e.g., by an AI assistant or for building other materials), not as a submission-ready resume.

Summary

Cybersecurity student at DePaul University with hands-on, self-directed experience across cloud security, detection engineering, DevSecOps, identity and access management, and full-stack web development. Builds complete, working, deployed projects rather than tutorials — each project below was designed, built, debugged, and documented independently, with a strong emphasis on understanding every line of code and every security decision rather than copy-pasting solutions.

Education

DePaul University — Chicago, IL (Loop Campus, CDM building) B.S. in Cybersecurity, Jarvis College of Computing and Digital Media Planned minor in Software Engineering (gated on passing CSC 300 — Data Structures) Status: Current student, 2026 - Present

College of DuPage — Glen Ellyn, IL **Associate in Applied Science (A.A.S.), Cybersecurity & Defense** — Completed GPA: 3.8 | Academic Honors (multiple terms) | Phi Theta Kappa Honor Society Transferred to DePaul after completing this degree.

Certifications earned at College of DuPage:

- Cybersecurity Specialist Certificate — Awarded with High Honors
- CIT CCNA Certificate — Awarded with High Honors

In progress:

- CompTIA Security+ (SY0-701) — actively studying
 - AWS Cloud Practitioner — actively studying via AWS Skill Builder
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Work Experience

Locker+ Associate — Amazon Downers Grove, IL October 2025 - August 2026

- Provided front-line customer and technical support, diagnosing and resolving on-site system and device issues in a fast-paced, high-volume environment.

- Trained and onboarded new hires on site systems, procedures, and daily workflows, explaining technical steps clearly to non-technical staff.
- Assisted management with computer and system tasks, device troubleshooting, and accurate record keeping and documentation.

Marketing & Operations Manager — Family Spa Business (incl. "One Spa") Managing marketing and day-to-day operations across three spa locations, including the launch of a new location. Also handles digital marketing (Google Ads, branding) for the family-owned salon Dhruva Brows Threading & Spa in Downers Grove, IL.

Projects — Full Detail

1. IAM Lifecycle Lab (Identity & Access Management)

Repo: `lakeshore-iam-lab` — github.com/Nsuthar1464/lakeshore-iam-lab **Platform/Tools:** Microsoft Entra ID (Azure Active Directory)

What it is: A simulated enterprise identity and access management environment built in Microsoft Entra ID, modeling the full employee lifecycle from hire to departure.

What was built:

- Implemented Role-Based Access Control (RBAC) and least-privilege access principles across a simulated organization.
- Modeled the complete Joiner-Mover-Leaver (JML) lifecycle: provisioning accounts for new hires, adjusting access as roles change, and fully deprovisioning access when someone leaves.
- Detected and remediated a simulated privilege escalation / unauthorized access incident using Entra ID audit logs — identified how an account gained access it shouldn't have had, traced it through the logs, and corrected the misconfiguration.
- Practiced user and group provisioning, access reviews, and permission auditing.

Why it matters: Demonstrates real identity governance skills — the exact skill set behind "access control & authentication" and "onboarding/offboarding" that shows up in nearly every IT support and security job description.

2. Cloud SOC Detection Lab (Security Operations Center / SIEM)

Repo: `sentinel-soc-detection-lab` — github.com/Nsuthar1464/sentinel-soc-detection-lab **Platform/Tools:** Microsoft Sentinel (cloud-native SIEM), KQL (Kusto Query Language), MITRE ATT&CK framework

What it is: A cloud Security Operations Center detection and monitoring lab built on Microsoft Sentinel, connected to Microsoft Entra ID as a live log source.

What was built:

- Deployed Microsoft Sentinel and connected it to Entra ID sign-in and audit logs as a data source.
- Wrote custom KQL (Kusto Query Language) queries to hunt through sign-in activity and audit logs for suspicious patterns, including extracting and parsing fields from nested JSON log data.
- Built a **scheduled analytics rule** mapped to the **MITRE ATT&CK framework** that automatically detects privilege escalation behavior and raises a security incident when triggered.
- Performed full incident triage end-to-end: assigning the incident, investigating root cause using the logs, and closing it out with a classification (true positive / benign / false positive).

Why it matters: This is hands-on SOC analyst / detection engineer work — the exact skillset for SOC Analyst, Security Analyst, and Detection Engineering roles. Directly demonstrates monitoring, detection, and incident response.

3. DevSecOps Pipeline (Infrastructure-as-Code Security Scanning)

Repo: `devsecops-terraform-pipeline` — github.com/Nsuthar1464/devsecops-terraform-pipeline **Platform/Tools:** Terraform (Infrastructure as Code), GitHub Actions (CI/CD), tfsec, Checkov, AWS

What it is: An automated CI/CD security pipeline that scans cloud infrastructure code for vulnerabilities before it can be deployed.

What was built:

- Built a GitHub Actions pipeline that automatically scans Terraform (Infrastructure-as-Code) files for AWS misconfigurations on every commit, using two scanning tools: **tfsec** and **Checkov**.
- Intentionally introduced real-world security flaws into the Terraform code to test the pipeline, including: publicly exposed S3 storage buckets, unrestricted SSH ingress rules (open to 0.0.0.0/0), and missing encryption on resources.
- Remediated each flaw with proper fixes: applied AWS KMS encryption, configured S3 public access blocks, and enforced least-privilege network security group rules.
- Configured the pipeline with both **advisory** (warns but doesn't block) and **blocking** (stops the deployment) scan gates, demonstrating risk-based security triage rather than a simple pass/fail gate.

Why it matters: This is real DevSecOps / cloud security engineering work — shifting security left into the development pipeline, which is a core practice at every modern cloud-first company (directly relevant to the AWS Cloud Security Engineer career goal).

4. Password Security Tool (Live, Deployed Web Application)

Repo: `password-security-tool` — github.com/Nsuthar1464/password-security-tool **Live**

URL: <https://nsuthar1464.github.io/password-security-tool/> **Platform/Tools:** HTML, CSS, JavaScript (vanilla, no frameworks), Web Crypto API, HaveIBeenPwned API

What it is: A fully deployed, live, client-side password security web application — the only project in the portfolio that is a complete shipped product with a real URL, not just a lab.

What was built — three integrated tools:

a) Password Strength Checker

- Real-time scoring of password strength (Weak / Medium / Strong) based on length, uppercase/lowercase variety, numbers, and symbols.

b) Password Generator (three modes)

- **Random mode:** Adjustable length slider (8-100 characters), toggleable numbers/symbols, and an optional base word that gets stylized with leetspice-style substitutions (a→@, i→1, e→3, o→0, s→\$) only when the relevant toggle is active.
- **Memorable mode:** Generates passphrases by joining random real words with dashes (e.g., "tiger-ocean-purple"), with a word-count slider and toggles for capitalization and full-word usage.
- **PIN mode:** Numeric-only codes with adjustable length and an optional custom-number seed (automatically strips any non-numeric characters).
- Includes copy-to-clipboard and refresh buttons; auto-regenerates on tab switch and slider changes.

c) Breach Checker (the technically most advanced feature)

- Checks whether a password has appeared in known data breaches using the **HaveIBeenPwned** public API.
- Implemented using **k-anonymity**: the password is hashed locally in the browser using SHA-1 via the **Web Crypto API**, and only the **first 5 characters** of that hash are sent to the API. The full hash and the real password **never leave the user's device**. The match against breach data happens locally, client-side.
- This was a deliberate privacy-first design choice — the explicit goal was that no third party, including the tool itself, ever sees the user's actual password.

Design: Apple-inspired visual design using the Inter font, three cards styled to look like macOS application windows (red/yellow/green traffic-light dots in the title bar), alternating left-right layout with supporting info panels beside each tool, scroll-triggered fade-in animations (built as a separate `animations.js` file), fully responsive layout that stacks correctly on mobile, and a footer crediting the creator with GitHub/LinkedIn links.

Deployment: Hosted live and free via GitHub Pages, fully tested in production (not just locally) — strength checker, all three generator modes, and the breach checker were all verified working on the live deployed version.

Why it matters: Demonstrates full front-end software engineering ability, an understanding of applied cryptography/privacy-preserving design (k-anonymity), API integration, and the ability to take a project from idea through build, debug, and live deployment — not just a tutorial or a local script.

5. Home Server / Self-Hosted Infrastructure Project

Hardware: Revived 2017 Dell Inspiron 3558 laptop (originally idle for years) **Platform/Tools:** Ubuntu Desktop LTS, UFW (firewall), fail2ban, Tailscale (VPN/mesh networking), Pi-hole (network-wide DNS ad-blocking), SSH

What it is: A self-built and self-hardened Linux home server project, approached deliberately with a "secure the device before deploying services on it" security-first methodology — security-by-design rather than bolted on afterward.

What was built:

- Revived old hardware: diagnosed and fixed a boot failure (fix was disabling Secure Boot in BIOS), installed Ubuntu Desktop LTS after evaluating and deciding against a more advanced distro (Omarchy/Arch Linux) as not appropriate for the hardware or the learning stage.
- **System hardening (completed before any services were deployed):**
 - Configured UFW firewall with a default-deny-incoming policy, explicitly allowing only necessary traffic.
 - Installed and enabled **fail2ban** to automatically detect and block brute-force login attempts.
 - Enabled automatic security updates (unattended-upgrades) so the system stays patched without manual intervention.
- **Remote access:** Configured SSH for remote administration, then installed and configured **Tailscale** (a mesh VPN) so the server can be securely accessed and administered from anywhere in the world — tested and confirmed working by connecting from an iPhone (via the Termius SSH client) over cellular data with home Wi-

Fi disabled, proving true location-independent access with nothing exposed to the public internet.

- **Lid-closed always-on operation:** Configured the laptop to continue running as a server with the lid closed (`systemd-logind` configuration), so it can operate tucked away like dedicated server hardware while still using its built-in battery as a natural UPS against power interruptions.
- **Network-level ad blocking:** Installed **Pi-hole** to function as a DNS-based, network-wide ad and tracker blocker (worked through a non-trivial installation process involving unattended-mode installation and firewall/port configuration to get it fully functional).
- **Planned/designed but intentionally deprioritized:** Full SSH key-based authentication (attempted; chose to prioritize working remote access over this particular hardening step given time constraints) and a planned authorized penetration-testing exercise using a personal Kali Linux VM against the hardened server, to validate the defenses that were put in place.

Why it matters: This project demonstrates real, hands-on Linux system administration, network security hardening, VPN/remote access architecture, and a security-first mindset applied to a real, physical, internet-connected system — not a classroom exercise. It is a genuine home-lab project of the kind valued in IT support, systems administration, and security engineering roles.

Technical Skills (Consolidated)

Cloud & SIEM: Microsoft Sentinel, Microsoft Entra ID, Azure, AWS (IAM, S3) fundamentals, KQL, Log Analytics, MITRE ATT&CK framework

DevSecOps & IaC: Terraform, GitHub Actions (CI/CD), tfsec, Checkov, static analysis / security scanning gates

Security & Identity: Incident response, detection engineering, IAM, RBAC, least-privilege design, access reviews, privilege escalation detection and remediation

Networking: TCP/IP, DNS, DHCP, VLANs, Cisco routing & switching (CCNA curriculum), Packet Tracer (built a 3-VLAN network with router-on-a-stick inter-VLAN routing and ACL-based security policies), network troubleshooting

Systems & Linux: Ubuntu/Linux system administration, UFW firewall configuration, fail2ban, SSH, Tailscale (VPN/mesh networking), Pi-hole/DNS filtering, systemd service configuration, Windows, macOS

Web Development: HTML, CSS, JavaScript (vanilla), Web Crypto API, REST API integration (HaveIBeenPwned), responsive design, GitHub Pages deployment

Tools & Platforms: Git/GitHub, Docker (containers, MCP server setup), VS Code, Claude Code, IntelliJ IDEA

Professional: Technical documentation and README writing, clear communication of technical concepts to non-technical audiences, customer service, fast learner, works well under pressure

Portfolio Links

- **GitHub:** github.com/Nsuthar1464
 - **Password Security Tool (live):** <https://nsuthar1464.github.io/password-security-tool/>
 - **LinkedIn:** [linkedin.com/in/sutharn555](https://www.linkedin.com/in/sutharn555)
 - Personal portfolio website: dark-themed, cybersecurity-focused, built with VS Code and Claude Code (not yet linked publicly in this document — add URL when ready)
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Document prepared as a full reference of all completed projects and experience, for use as source material. All details reflect actual work completed; nothing in this document is invented or placeholder.